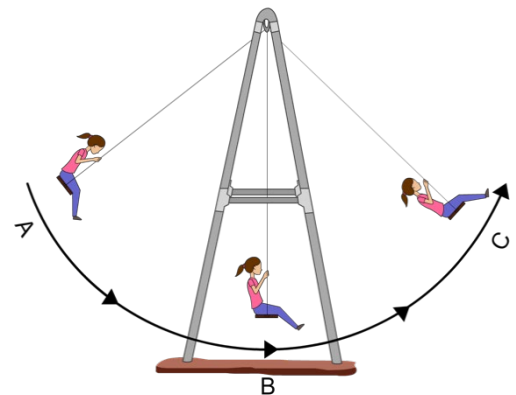


Student Instructions:

- Write your complete ID code on your answer sheet.
 - If you change your answer, erase well.
1. Vinegar, air, and saltwater are examples. Which choice below is a homogeneous mixture of two substances that retain their chemical properties?
 - A. alloy
 - B. solution
 - C. suspension
 - D. colloid
 2. In which one of the following processes does a solid change directly into a gas without passing through a liquid phase?
 - A. sublimation
 - B. transpiration
 - C. vaporization
 - D. condensation
 3. Three of the choices below are mixtures. Which one is a compound?
 - A. mud
 - B. smog
 - C. cement
 - D. table salt
 4. The following four forces affect flight. Which one is the force of gravity that pulls a flying object toward Earth's center?
 - A. drag
 - B. thrust
 - C. lift
 - D. weight
 5. Examine the diagram of a child on a swing. At which position is the swinging child's potential energy LOWEST?
 - A. A
 - B. B
 - C. C
 - D. The potential energy is the same in all 3 positions.
 6. Below are four methods of heat transfer. Which one occurs between objects in contact with each other, such as a pan and the food cooking inside it?
 - A. conduction
 - B. convection
 - C. radiation
 - D. advection



7. The equation seen here states Ohm's Law, where V is voltage in volts, I is current in amperes, and R is resistance in ohms. According to this law, what is the voltage of a circuit with a current of 2 amps and a resistance of 13 ohms?

$$V = IR$$

- A. 2 volts
- B. 6.5 volts
- C. 13 volts
- D. 26 volts

8. Which choice below is NOT a type of coal?

- A. anthracite
- B. bituminous
- C. fluorite
- D. lignite

9. Below are four soil layers. Which one is both the deepest and hardest?

- A. topsoil
- B. bedrock
- C. subsoil
- D. parent material

10. Match each fossil category with its example.

- x. true form
- y. trace
- z. cast

- 1. dinosaur skull
- 2. footprint
- 3. filled mold of a trilobite

- A. x-1, y-2, z-3
- B. x-2, y-3, z-1
- C. x-1, y-3, z-2
- D. x-2, y-1, z-3

11. They are dense and have ragged edges. Which of the following categories encompasses the clouds that can most often cause rain, snow, hail, and even tornadoes?

- A. cirrus
- B. cumulus
- C. nimbus
- D. stratus

12. Which gas below composes approximately 78% of Earth's atmosphere?

- A. carbon dioxide
- B. oxygen
- C. nitrogen
- D. argon

13. Triton is one of its 14 moons. Which one of the following gas giants can't be seen with the naked eye?

- A. Neptune
- B. Saturn
- C. Jupiter
- D. Uranus

14. Below are four lunar phases. During which one is the lunar disk invisible to the naked eye, making for the darkest sky of the month?
- A. waning moon
 - B. full moon
 - C. waxing moon
 - D. new moon
15. Because they are catalysts, they control how fast chemical reactions occur. Which of the following choices names these substances that can aid digestion and growth?
- A. fissions
 - B. enzymes
 - C. flagella
 - D. methanogens
16. Below are four parts of a cell. Which one is made of RNA and is responsible for manufacturing proteins?
- A. mitochondrion
 - B. nucleoplasm
 - C. ribosome
 - D. vacuole
17. Which of the following statements about archaea is FALSE?
- A. They perform photosynthesis.
 - B. They are prokaryotes.
 - C. They reproduce asexually.
 - D. They are unicellular.
18. Which one of the following plants is nonvascular?
- A. fern
 - B. bamboo
 - C. bluegrass
 - D. moss
19. Three of the choices below are parts of the pistil, the flower's female organ. Which one is part of the stamen?
- A. stigma
 - B. anther
 - C. style
 - D. ovary
20. Match each invertebrate with its phylum.
- | | |
|---------------|------------------|
| x. octopus | 1. Echinodermata |
| y. jellyfish | 2. Cnidaria |
| z. sea urchin | 3. Mollusca |
- A. x-2, y-3, z-1
 - B. x-1, y-2, z-3
 - C. x-3, y-2, z-1
 - D. x-2, y-1, z-3

21. Which one of the following choices is NOT a role in a beehive?
- A. queen
 - B. worker
 - C. drone
 - D. grader
22. Three of the animals below are cold-blooded. Which one is warm-blooded?
- A. killer whale
 - B. rattlesnake
 - C. salmon
 - D. red-eyed tree frog
23. Which of the following species is a primate?
- A. squirrel
 - B. lemur
 - C. parrot
 - D. crocodile
24. The trachea, bronchial tubes, larynx, and pharynx are all part of which human body system?
- A. nervous
 - B. respiratory
 - C. lymphatic
 - D. endocrine
25. A virus causes which one of the following illnesses, whose technical name is varicella?
- A. salmonella
 - B. strep throat
 - C. tuberculosis
 - D. chicken pox
26. What piece of lab equipment is seen here?
- A. beaker
 - B. crucible
 - C. Erlenmeyer flask
 - D. pipette
27. Which choice below names the aspects of a scientific experiment that are kept constant?
- A. controls
 - B. independent variables
 - C. tropisms
 - D. dependent variables
28. It can be mercury or aneroid. Which of the following weather station tools measures atmospheric pressure?
- A. barometer
 - B. hygrometer
 - C. anemometer
 - D. thermometer
29. Jane Goodall spent most of her career studying what animals in their natural habitats?
- A. gorillas
 - B. orangutans
 - C. chimpanzees
 - D. spider monkeys



30. Match each man with the disease whose vaccine he developed.

- x. Edward Jenner
- y. Jonas Salk
- z. Louis Pasteur

- 1. polio
- 2. smallpox
- 3. anthrax

- A. x-1, y-3, z-2
- B. x-3, y-2, z-1
- C. x-2, y-3, z-1
- D. x-2, y-1, z-3

TIEBREAKERS 31-36

31. They are up to 1,000 times brighter than the average galaxy. Which choice below names these objects that are among the most distant known objects in the universe?

- A. meteorites
- B. quasars
- C. novas
- D. pulsars

32. Which one of the following plants is a gymnosperm?

- A. rose
- B. hemp
- C. apple tree
- D. ginkgo

33. Which stage below begins both complete and incomplete metamorphosis?

- A. pupa
- B. larva
- C. egg
- D. imago

34. Which one of the following animals is a carnivore?

- A. coyote
- B. goat
- C. camel
- D. elephant

35. Below are four words related to wave behavior. In which one does a wave change direction when moving from one medium to another, as from the air to a glass of water?

- A. reflection
- B. diffraction
- C. refraction
- D. polarization

36. Which of the following choices is NOT one of Benjamin Franklin's inventions?

- A. bifocals
- B. telegraph
- C. lightning rod
- D. glass harmonica

**Science – Elementary – Regional
2023 Governor’s Cup**

1. B
2. A
3. D
4. D
5. B
6. A
7. D
8. C
9. B
10. A
11. C
12. C
13. A
14. D
15. B
16. C
17. A
18. D
19. B

20. C
21. D
22. A
23. B
24. B
25. D
26. C
27. A
28. A
29. C
30. D
31. B
32. D
33. C
34. A
35. C
36. B

TIEBREAKERS 31-36

Notes for Inquiry Panel

[illegible]

1: 1, 2, 3
2: 4, 5
3: 6, 7, 35
4: 8, 9, 10
5: 11, 12
6: 13, 14, 31
7: 15, 16, 17
8: 18, 19, 32
9: 20, 21, 33
10: 22, 23, 34
11: 24, 25
12: 26, 27, 28
13: 29, 30, 36